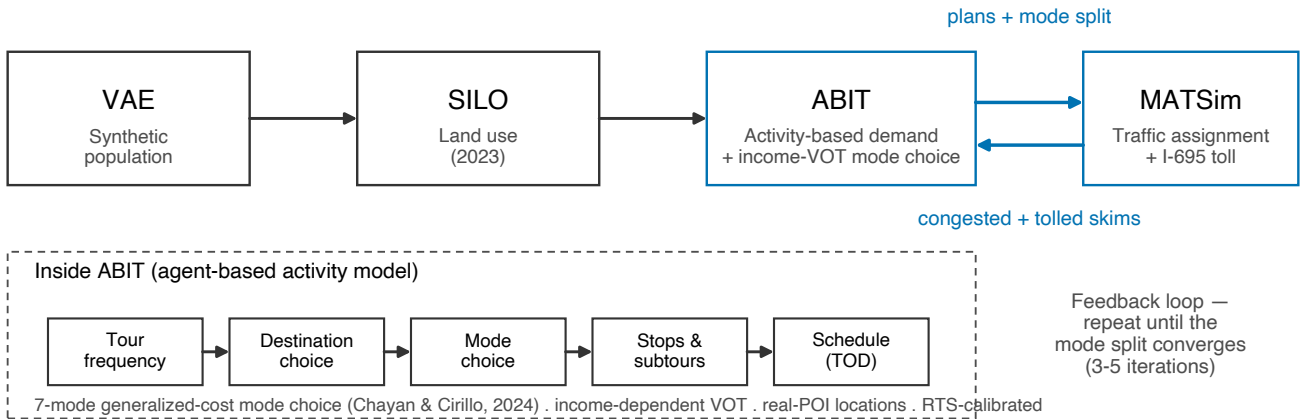




Feedback-loop algorithm

- 0 ABIT builds base daily plans from SILO / VAE — tours, POI-based activity locations, income-VOT mode choice on free-flow skims.
- 1 MATSim assigns the plans on the tolled network; its inner loop re-routes and re-times trips (modes held fixed).
- 2 Extract updated origin-destination skims from MATSim: congested car travel time + per-OD toll cost.
- 3 ABIT re-runs its income-VOT generalized-cost mode choice on the new skims to produce an updated mode split.
- 4 If the mode split changed by less than the tolerance, stop; otherwise return to Step 1.

Outputs: mode shift . toll revenue . consumer surplus (equity) by income / race / age . departure-time shift.

Figure. Coupled VAE -> SILO -> ABIT <-> MATSim pipeline. ABIT (activity-based demand with income-VOT mode choice) and MATSim (traffic assignment with the I-695 toll) iterate: ABIT sends plans + mode split, MATSim returns congested, tolled skims, and the loop repeats until the mode split converges.